Vulnerability Assessment & Penetration Testing (VAPT) Lab Report

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Date: 02 March 2026

Environment: Virtual Lab (Host-Only / Internal Network)

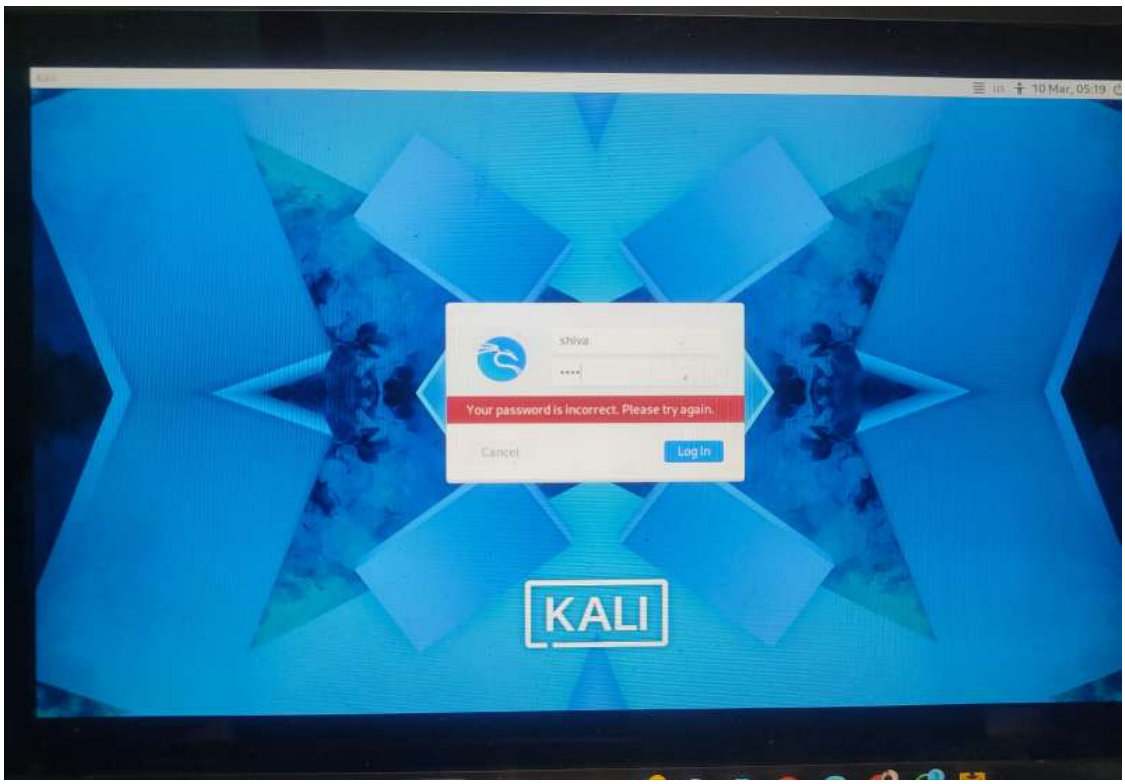
WEEK 1 – Lab Setup & Basic Scanning

1. Virtual Lab Setup

A virtual lab environment was configured using a hypervisor (VirtualBox/VMware). The network mode was set to Host-Only or Internal Network to ensure isolation from the production network. This configuration allows safe vulnerability testing.

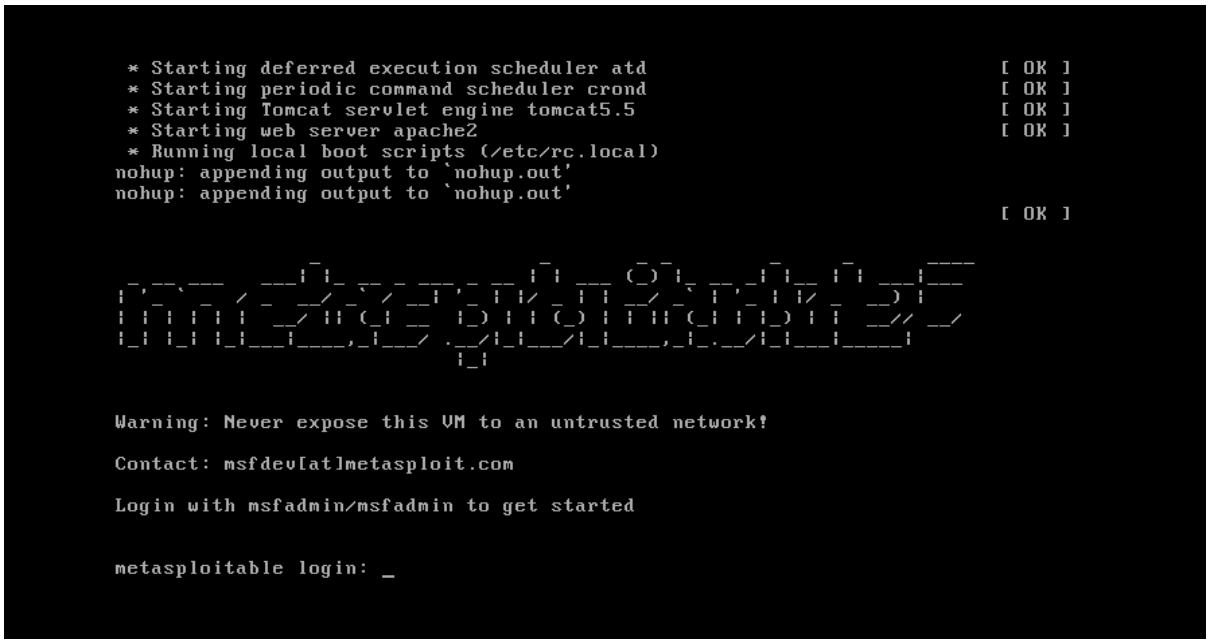
2. Attacker Machine – Kali Linux

Kali Linux was installed as the attacker machine. After installation, the system was updated using 'sudo apt update && sudo apt upgrade -y'. Tools verified: Nmap and OpenVAS (Greenbone).



3. Target Machines Installed

• Metasploitable 2 (Intentionally vulnerable Linux VM)



Login with default credentials

Username: msfadmin

Password: msfadmn

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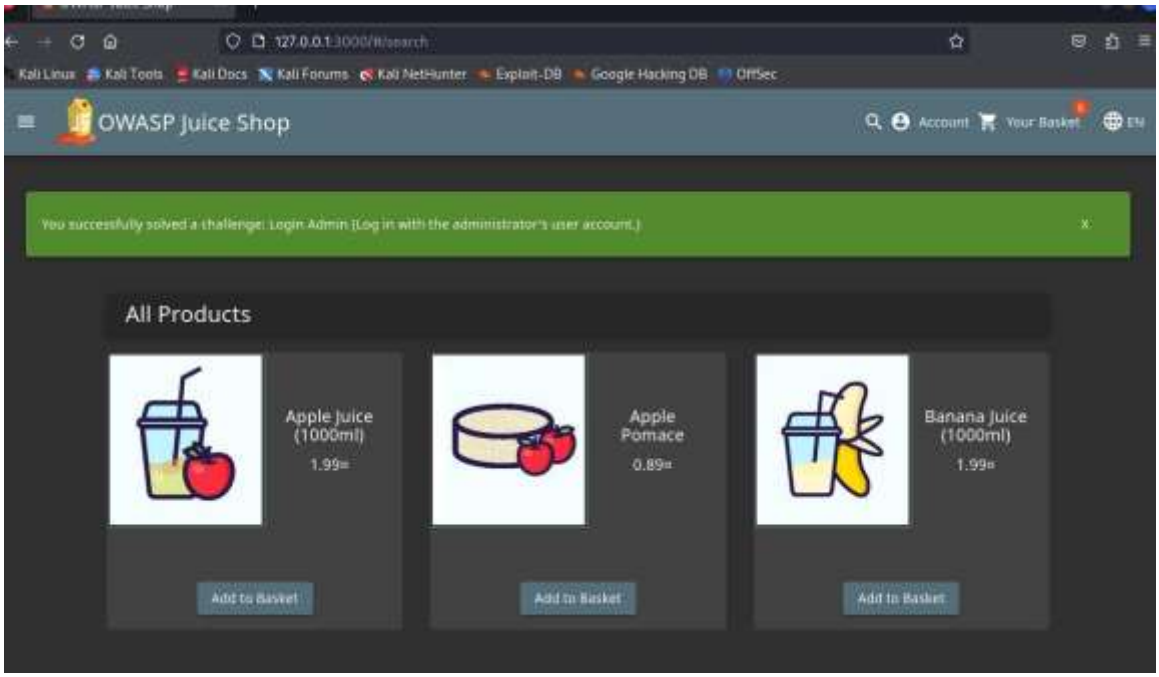
To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 00:0c:29:ad:2e:80
          inet addr:192.168.230.131  Bcast:192.168.230.255  Mask:255.255.255.0
          inet6 addr: fe80::20c:29ff:fead:2e80/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:50 errors:0 dropped:0 overruns:0 frame:0
          TX packets:64 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:4999 (4.8 KB)  TX bytes:7008 (6.8 KB)
          Base address:0x2000 Memory:fd5c0000-fd5e0000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:97 errors:0 dropped:0 overruns:0 frame:0
          TX packets:97 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:21529 (21.0 KB)  TX bytes:21529 (21.0 KB)

msfadmin@metasploitable:~$
```

IP = 192.168.230.131

• OWASP Juice Shop (Deployed using Docker on port 3000)



4. Basic Scanning with Nmap

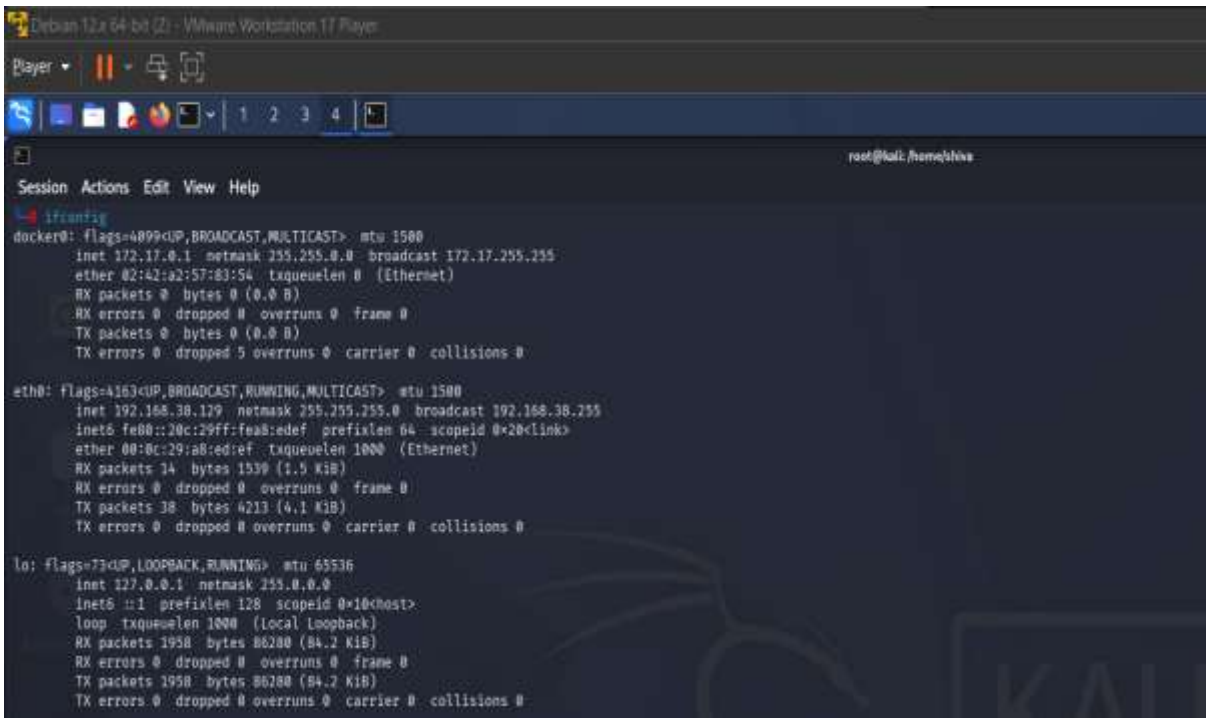
Initial reconnaissance was performed using Nmap to identify live hosts, open ports, running services, and operating system details.

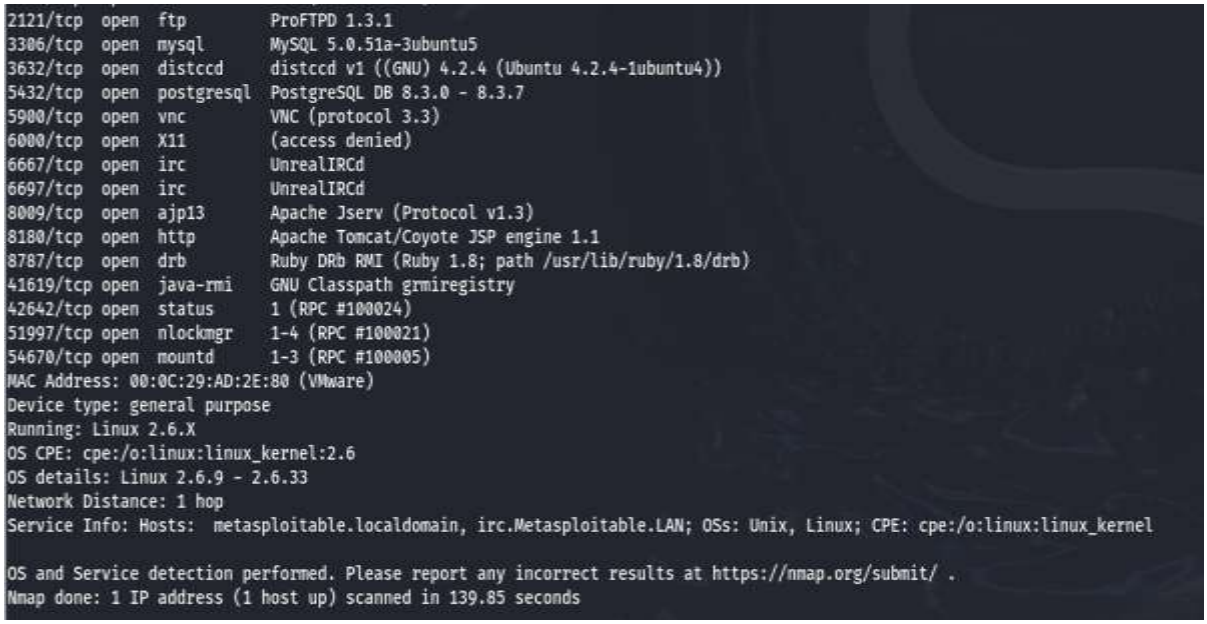
Commands Used:

nmap -sS <target-ip>

nmap -sV <target-ip>

nmap -A -p- <target-ip>





WEEK 2 – Vulnerability Assessment & Documentation

1. Network & Port Scanning

Network discovery was performed to identify active systems within the lab environment. Port enumeration and OS fingerprinting were conducted.

Commands Used:

nmap -sn <network-range>

nmap -sV -O <target-ip>

Insert Network Scan screenshots below:

```
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

┌──(root㉿kali)-[/home/shiva]
└─# nmap -sn 192.168.38.129
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-11 07:11 -0400
Nmap scan report for 192.168.38.129
Host is up.
Nmap done: 1 IP address (1 host up) scanned in 0.50 seconds

┌──(root㉿kali)-[/home/shiva]
└─# nmap -sS -p- 192.168.38.129
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-11 07:12 -0400


┌──(root㉿kali)-[/home/shiva]
└─# nmap --script vuln 192.168.38.129
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-11 07:24 -0400
Nmap scan report for 192.168.38.129
Host is up.
All 1000 scanned ports on 192.168.38.129 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 211.97 seconds

┌──(root㉿kali)-[/home/shiva]
└─#
```

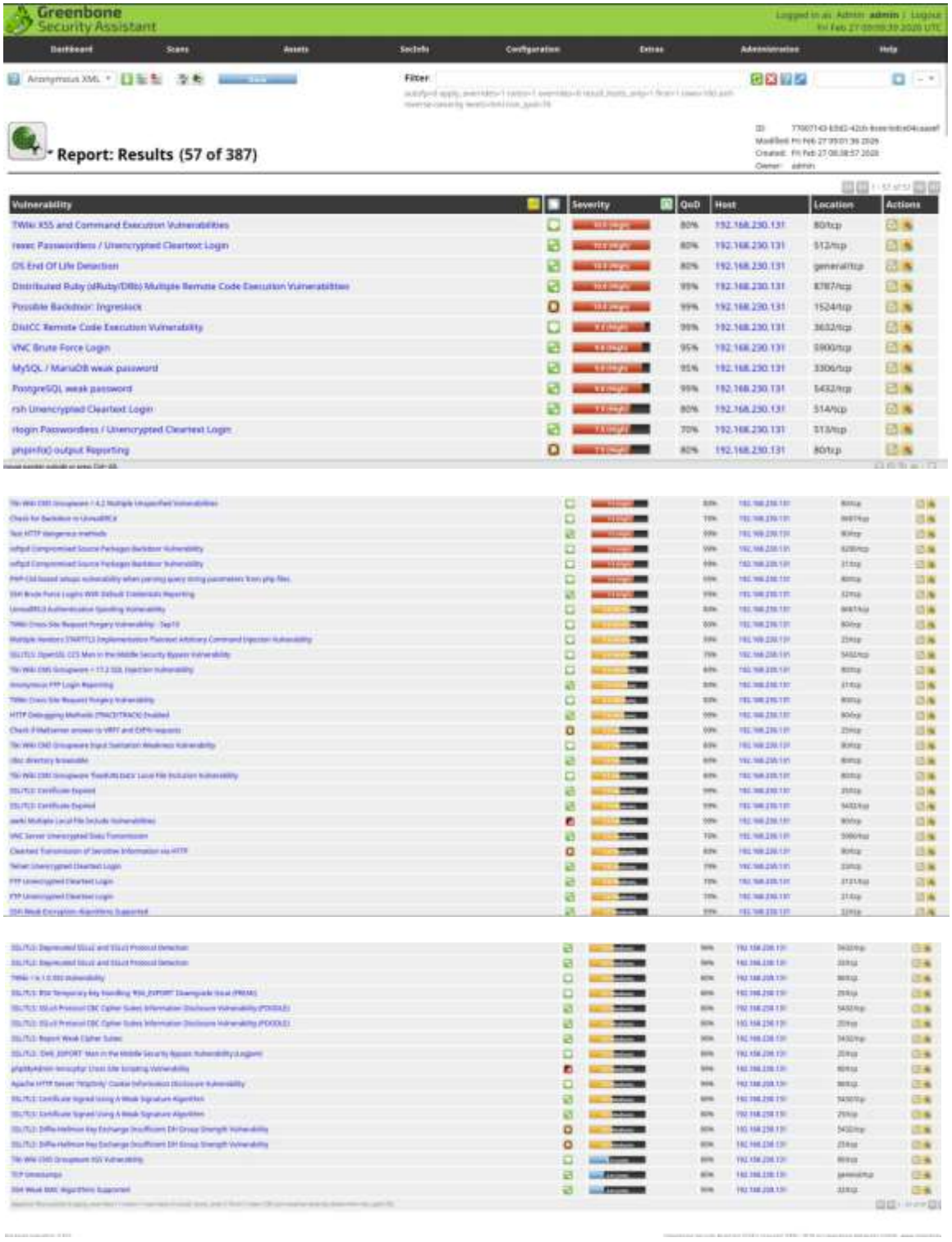
```
2121/tcp  open  ftp         ProFTPD 1.3.1
3306/tcp  open  mysql       MySQL 5.0.51a-3ubuntu5
3632/tcp  open  distccd     distccd v1 ((GNU) 4.2.4 (Ubuntu 4.2.4-1ubuntu4))
5432/tcp  open  postgresql  PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc         VNC (protocol 3.3)
6000/tcp  open  X11         (access denied)
6667/tcp  open  irc         UnrealIRCd
6697/tcp  open  irc         UnrealIRCd
8009/tcp  open  ajp13       Apache Jserv (Protocol v1.3)
8180/tcp  open  http        Apache Tomcat/Coyote JSP engine 1.1
8787/tcp  open  drb         Ruby DRb RMI (Ruby 1.8; path /usr/lib/ruby/1.8/drb)
41619/tcp open  java-rmi    GNU Classpath grmiregistry
42642/tcp open  status      1 (RPC #100024)
51997/tcp open  nlockmgr    1-4 (RPC #100021)
54670/tcp open  mountd      1-3 (RPC #100005)
MAC Address: 00:0C:29:AD:2E:80 (VMware)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop
Service Info: Hosts:  metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 139.85 seconds
```

2. Vulnerability Assessment using OpenVAS

OpenVAS (Greenbone Vulnerability Manager) was used to perform automated vulnerability scanning against Metasploitable. The 'Full and Fast' scan configuration was selected.

3. Identified Vulnerabilities : 57 vulnerbilities found



4. Risk Classification

Vulnerabilities were classified based on CVSS scoring standards. Critical and High severity findings require immediate remediation, while Medium and Low findings should be addressed through security hardening practices.

5. Conclusion

The vulnerability assessment successfully identified multiple security weaknesses within the lab environment. This exercise enhanced understanding of network enumeration, vulnerability identification, CVE mapping, and risk assessment. Proper patch management and configuration hardening are recommended.